

Computational Microelectronics

Lecture 2

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L2

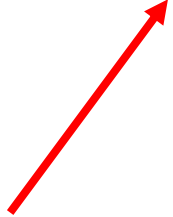
Kirchhoff's current law

- OUT node

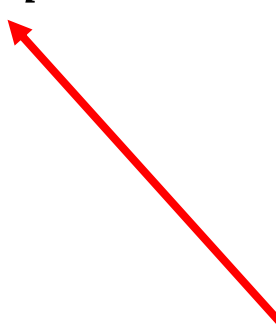
- Sum of two branch currents must vanish:

$$I_{Dn}(V_{in}, V_{out}) + I_{Dp}(V_{in}, V_{out}) = 0$$

A nonlinear function



Another nonlinear function



- Nonlinearity is the root cause of difficulties.

Counterexample

- Consider a linear system, instead of a nonlinear system.
 - The KCL reads

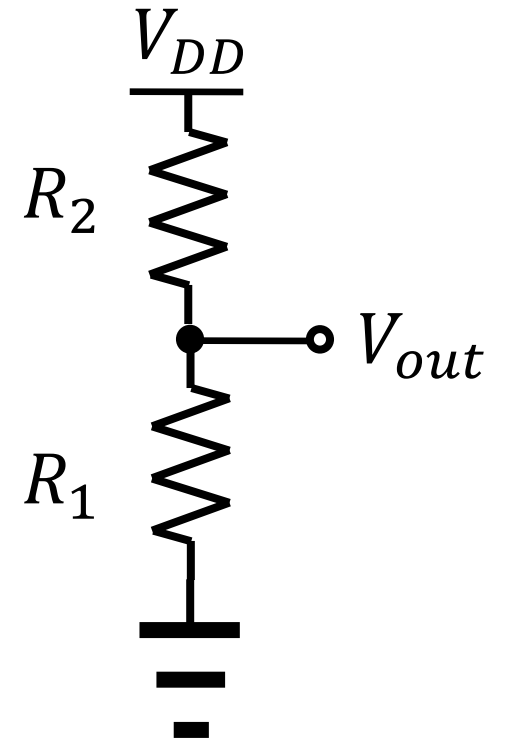
$$I_{R1}(V_{out}) = I_{R2}(V_{out})$$

- For resistors, Ohm's law must be applied.

$$\frac{V_{out}}{R_1} = \frac{V_{DD} - V_{out}}{R_2}$$

- Solution can be found immediately

$$V_{out} = \frac{R_1}{R_1 + R_2} V_{DD}$$



Discussion

- Remember our previous lecture.
 - When V_{in} is 0.5 V and V_{out} is 0.7 V, we have $I_{Dn} = 1.77 \mu\text{A}$ and $I_{Dp} = -7.92 \mu\text{A}$.
 - In L1, we increased V_{out} by 0.1 V.
 - Q) Why did we increase it?
 - A) In order to balance two currents. (We know that I_{Dn} must be increased.)
 - Q) Why 0.1 V?
 - A) Just heuristically.
 - Can we find a better way to update V_{out} ?

Our approach

- With an initial set of V_{in} and V_{out} , the KCL (usually) does not hold.

$$I_{Dn}(V_{in}, V_{out}) + I_{Dp}(V_{in}, V_{out}) \neq 0$$

- We assert that a better solution, $V_{out} + \delta V_{out}$, satisfies the KCL.

$$I_{Dn}(V_{in}, V_{out} + \delta V_{out}) + I_{Dp}(V_{in}, V_{out} + \delta V_{out}) = 0$$

- Now, we want to find an equation for δV_{out} .
- The exact equation for δV_{out} is still very difficult to solve.
- An appropriate approximation is needed.

Linear approximation

- Key step toward a practical method

- An exact equation

$$I_{Dn}(V_{in}, V_{out} + \delta V_{out}) + I_{Dp}(V_{in}, V_{out} + \delta V_{out}) = 0$$

- An approximate equation

$$I_{Dn}(V_{in}, V_{out}) + \left. \frac{\partial I_{Dn}}{\partial V_{out}} \right|_{V_{in}, V_{out}} \delta V_{out} + I_{Dp}(V_{in}, V_{out}) + \left. \frac{\partial I_{Dp}}{\partial V_{out}} \right|_{V_{in}, V_{out}} \delta V_{out} \approx 0$$

Calculation of update

- After simple manipulation,
 - An approximate equation for δV_{out} can be rearranged as:

$$\left[\left. \frac{\partial I_{Dn}}{\partial V_{out}} \right|_{V_{in}, V_{out}} + \left. \frac{\partial I_{Dp}}{\partial V_{out}} \right|_{V_{in}, V_{out}} \right] \delta V_{out} = - \left[I_{Dn}(V_{in}, V_{out}) + I_{Dp}(V_{in}, V_{out}) \right]$$

- Two terms:
 - Value of $I_{Dn} + I_{Dp}$ evaluated at the present solution set
 - Sensitivity of $I_{Dn} + I_{Dp}$ with respect to change of solution

Case) Saturation region

- Function value

- It is straightforward to evaluate the currents:

$$I_{Dn} = \frac{KP_n}{2} \frac{W_{effn}}{L_{effn}} (1 + LAMBDA_n \times V_{out}) (V_{in} - V_{tn})^2$$

$$I_{Dp} = -\frac{KP_p}{2} \frac{W_{effp}}{L_{effp}} \left(1 - LAMBDA_p \times (V_{out} - V_{DD})\right) (V_{in} - V_{DD} - V_{tp})^2$$

- (We can derive similar expressions in the linear region.)

Case) Saturation region

- Sensitivity

- Partial derivative with respect to V_{out} is taken:

$$\frac{\partial I_{Dn}}{\partial V_{out}} = \frac{KP_n}{2} \frac{W_{effn}}{L_{effn}} LAMBDA A_n (V_{in} - V_{tn})^2$$

$$\frac{\partial I_{Dp}}{\partial V_{out}} = \frac{KP_p}{2} \frac{W_{effp}}{L_{effp}} LAMBDA A_p (V_{in} - V_{DD} - V_{tp})^2$$

- (We can derive similar expressions in the linear region.)

Numeric values

- When $V_{in} = 0.5$ V, the answer for V_{out} is about 1.157566 V.
 - Let us assume that we start from $V_{out} = 1.1$ V.
 - Then, the function value is -2.189 μ A.
 - The sensitivity is 33.75 μ A/V.
 - The calculated update is 0.064859 V.
- Try again with $V_{out} = 1.164859$ V.
- The calculated update is -0.007204 V.
- Try again with $V_{out} = 1.157655$ V.
- The calculated update is -0.000089 V.

Newton method

- We want to solve the following equation:

$$f(x) = 0$$

- Assume an approximate solution, x_n .
- Evaluate the function value, $f(x_n)$. (It should vanish, but it does not.)
- Evaluate the sensitivity, $\left. \frac{df}{dx} \right|_{x_n}$.
- Calculate the update, $\delta x = -f(x_n) / \left(\left. \frac{df}{dx} \right|_{x_n} \right)$.
- Repeat the above procedure with $x_{n+1} = x_n + \delta x$.

Application of Taylor's theorem

- True solution is α .

$$f(\alpha) = 0$$

- By using Taylor's theorem,

$$f(\alpha) = f(x_n) + \left. \frac{df}{dx} \right|_{x_n} (\alpha - x_n) + \frac{1}{2} \left. \frac{d^2 f}{dx^2} \right|_{\xi_n} (\alpha - x_n)^2$$

(Note that ξ_n is between x_n and α .)

- After simple manipulation,

$$x_n - f(x_n) / \left(\left. \frac{df}{dx} \right|_{x_n} \right) - \alpha = \frac{1}{2} \left(\left. \frac{d^2 f}{dx^2} \right|_{\xi_n} \right) / \left(\left. \frac{df}{dx} \right|_{x_n} \right) (x_n - \alpha)^2$$

Quadratic convergence

- The LHS can be further simplified.
 - Remember that

$$x_{n+1} = x_n + \delta x = x_n - f(x_n) / \left(\left. \frac{df}{dx} \right|_{x_n} \right)$$

- Therefore,

$$x_{n+1} - \alpha = \frac{1}{2} \left(\left. \frac{d^2 f}{dx^2} \right|_{\xi_n} \right) / \left(\left. \frac{df}{dx} \right|_{x_n} \right) (x_n - \alpha)^2$$

- Quadratic convergence

Caution for practical application

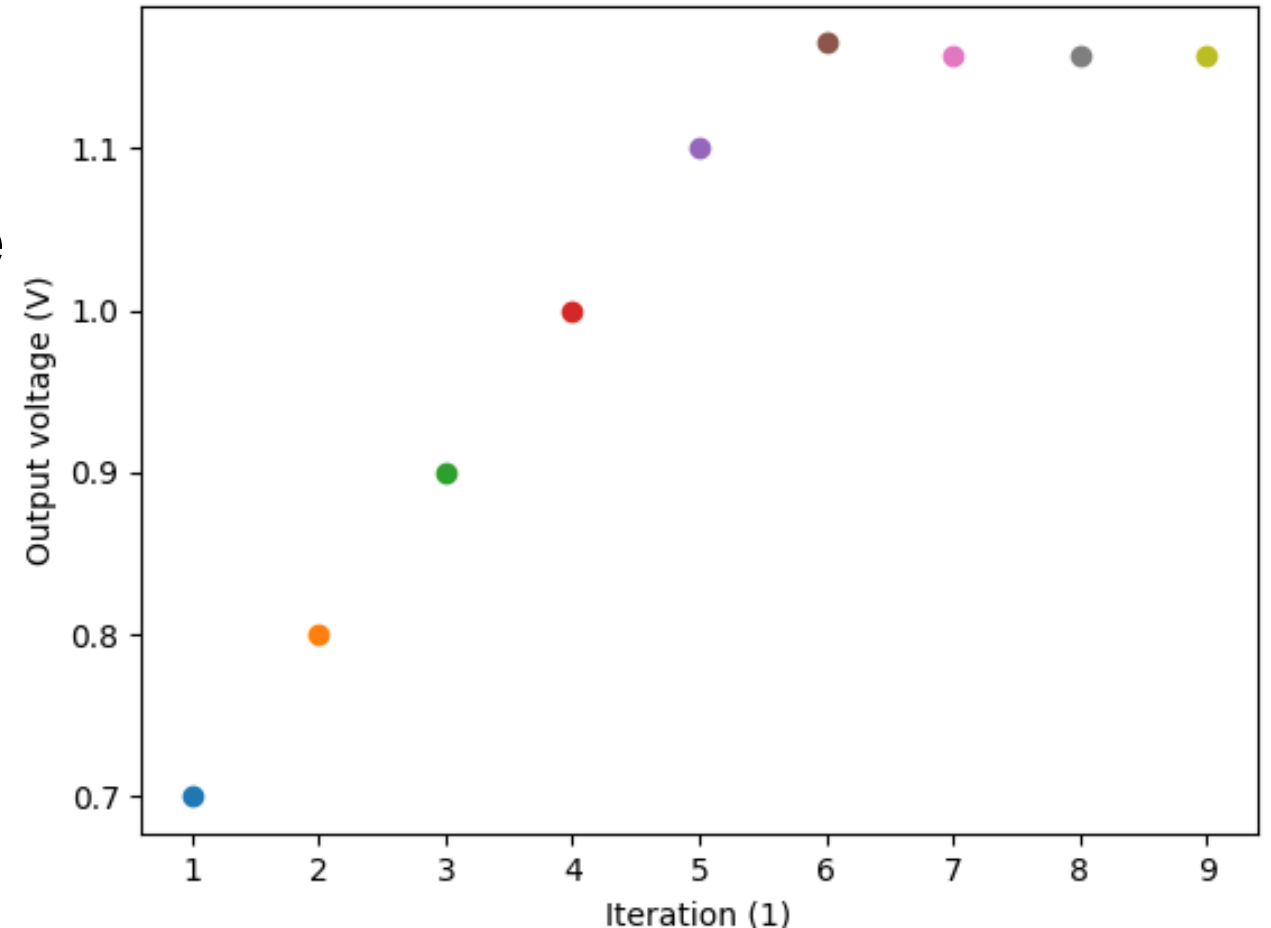
- The Newton method is sensitive to x_n .
 - Let us assume that we start from $V_{out} = 1.0$ V. (Still, $V_{in} = 0.5$ V)
 - Then, the update becomes 0.263083 V.
 - It suggests that our next solution is 1.263083 V, which exceeds V_{DD} .
- It is very important to start from a reasonable initial solution.
 - When the update is too large, it should be damped.
 - For example, we can limit the (absolute) update up to 0.1 V.
 - Also, V_{out} is bounded within $[0, V_{DD}]$.

Number of iterations

- When we have a voltage change smaller than 1 μV , the calculation stops.

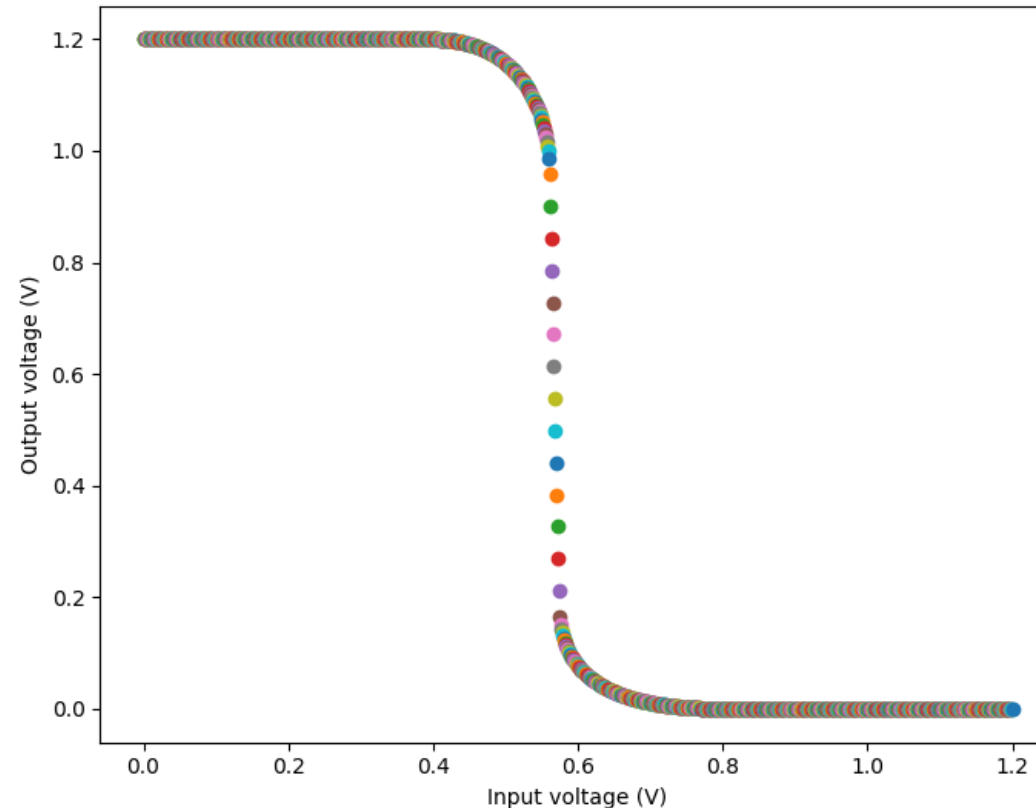
- Start at $\frac{V_{DD}}{2}$.
- Maximum (absolute) change is limited to 0.1 V.
- 9 iterations at $V_{in} = 0.5 \text{ V}$

```
1 0.7 0.1
2 0.7999999999999999 0.1
3 0.8999999999999999 0.1
4 0.9999999999999999 0.1
5 1.0999999999999999 0.1
6 1.1648592592592593 0.06485925925925938
7 1.1576546605493248 -0.007204598709934521
8 1.1575659813078383 -8.86792414864831e-05
9 1.1575659678253083 -1.3482529929426225e-08
```



Homework#2

- Due: AM08:00, September 14
- Problem#1
 - Calculate the voltage transfer curve.



Thank you!